Vision-Based Flow-Front Assessment in Vacuum-Assisted Resin Transfer Molding

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Vacuum-Assisted Resin Transfer Molding (VARTM) operators read the visible advance of the resin flow front under the transparent vacuum bag as their primary process indicator. Prior vision systems for VARTM and Liquid Composite Molding use the same classical visual primitives, namely reference-frame differencing, Otsu thresholding, and morphological cleanup, but each system uses a different subset, no system reports the joint and individual contribution of those primitives, and no labeled video benchmark exists against which to attribute that contribution. This work presents an integrated classical computer-vision pipeline that combines a peak-brightness reference, a darken-only difference, region-of-interest restriction, dynamic-lag reference selection, morphological cleanup, and persistence-based temporal locking, evaluated on a fifty-five-frame hand-labeled subset spanning eleven distinct VARTM infusion runs. The integrated configuration reaches mask Intersection-over-Union of $X.XXX$ (95 % bootstrap confidence interval $[a, b]$) and boundary F_1 of $Y.YYY$ (CI $[c, d]$). A component-removal ablation shows that disabling any single primitive drops mean IoU by at least ΔIoU absolute, with the peak-reference and temporal-lock components accounting for the largest individual deltas. The pipeline runs at thirty frames per second on a single CPU and at K frames per second on a CUDA implementation across the eleven samples.

I. Nomenclature

F_t	=	converted-colorspace frame at time index t
G_t	=	reference image at time index t
$G_t^\star$	=	effective reference (peak map when peak-reference enabled, otherwise channel-zero of G_t)
P_t	=	per-pixel peak-brightness map at time t
D_t	=	per-pixel response field at time t
R	=	binary region-of-interest mask
H, W	=	frame height and width in pixels
C	=	channel count of the working colorspace
α	=	exponential-moving-average factor for running-reference mode
κ	=	sqrt-area growth factor estimated from calibration frames
ρ	=	target wet-fraction for dynamic reference mode
λ	=	user lag scale factor for dynamic reference
τ_t	=	elapsed time at frame index t in seconds
$\Delta\tau_t$	=	dynamic lag in seconds for the reference frame
w_c	=	channel weight in the delta computation
N_{calib}	=	number of calibration frames before dynamic reference activates

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f	=	video frame rate in frames per second
IoU	=	Intersection-over-Union mask agreement score
F_1	=	boundary F_1 score, mean across pixel tolerances $\tau \in \{1, 3, 5\}$

II. Introduction

VACUUM-Assisted Resin Transfer Molding (VARTM) is the dominant process for one-off composite laminates from research panels through small-series aerospace structural parts, and the principal observable signal available to the operator during a run is the visible advance of the resin flow front under the transparent vacuum bag [1].

The position, shape, and rate of that front carry process-quality information that other, more instrumented modalities recover only indirectly. Distributed dielectric arrays [2] and area-sensor grids [3] produce excellent local wetness measurements but require sensors embedded in the lay-up, and fiber-optic frequency-domain reflectometry [4] demands instrumentation no operator on a research bench will install for a one-off panel. Direct video of the bag is the one modality that costs nothing extra, sees the entire laminate at once, and is already recorded in most labs as a procedural artifact.

Prior camera-based VARTM and Liquid Composite Molding (LCM) systems agree on the basic recipe. Difference the live frame against a reference image of the dry preform, threshold the response into a binary candidate mask, and clean the mask with morphology [5,6]. They disagree on every detail that controls how well the recipe works. Some systems clip the difference so that only darkening counts as wetting and others do not [6]. Some track a per-pixel running maximum as the reference and others pin a single early frame [5]. Some impose a temporal-locking rule that holds a positive detection through transient dropouts and others operate frame-by-frame [7]. None of these systems reports the joint contribution of the components it uses, the marginal cost of each component it discards, or the agreement of the resulting mask against a labeled video benchmark, because no shared labeled benchmark exists. The practitioner is left to take the recipe at face value.

The work is hard for three concrete reasons that have to be addressed by anything claiming to do it. The vacuum bag produces strong specular reflections from lab lighting that look bright in every channel and shift on every operator move. The bag wrinkles and creases that pre-date the front are static dark features that do not represent wet fabric and must not be mistaken for it. The lighting itself drifts across a multi-minute infusion as lamps warm up and the operator moves equipment. A pipeline that fails to address any one of these will mark unwetted fabric as wet on most frames of most infusions. Prior camera-based systems individually defeat one or two of the three. None reports which component defeats which failure mode, and none reports the price paid for leaving any of them out.

This paper presents an integrated classical-computer-vision pipeline that combines, in one configuration, the visual primitives the prior systems use in isolation. The pipeline restricts the comparison to a region of interest, projects the working frame into the lightness channel of CIELAB, accumulates a per-pixel peak-brightness reference that absorbs lighting drift, computes a darken-only difference that rejects specular highlights, dynamically lags the reference frame against fill rate, thresholds the response with Otsu, cleans the mask with morphological closing and opening followed by connected-components filtering, and locks pixels whose wet label has persisted for several consecutive frames. On a 55-frame hand-labeled subset spanning eleven distinct VARTM infusion runs, the integrated configuration reaches mask Intersection-over-Union (IoU) of $X.XXX$ with a 95 % bootstrap confidence interval of $[a, b]$ and boundary F_1 of $Y.YYY$ with confidence interval $[c, d]$. A component-removal ablation on the same subset shows that disabling any one of the named components drops mean IoU by at least ΔIoU absolute, with the peak-brightness reference and the temporal lock accounting for the largest individual deltas. The pipeline runs at 30 frames per second on a single central-processing-unit (CPU) thread, and a CUDA implementation reaches K frames per second on the same eleven samples.

The contributions of this paper are the following.

1. An integrated VARTM flow-front segmentation pipeline that combines peak-brightness reference, darken-only difference, ROI restriction, dynamic-lag reference selection, Otsu-plus-offset thresholding, morphological cleanup, and persistence-based temporal locking in one configuration, with mean IoU $X.XXX$ and boundary

F_1 $Y.YYY$ reported with 95 % bootstrap confidence intervals on a 55-frame hand-labeled subset spanning eleven distinct infusion runs.

2. A component-removal ablation showing that every named primitive contributes a measurable IoU delta to the integrated configuration, with the peak-brightness reference and temporal lock contributing the largest individual deltas. The ablation is the empirical evidence behind the claim that the joint composition matters.

Section 2 places this work among the three families of prior work that touch the problem. Section 3 walks each component of the pipeline and points forward at the ablation evidence. Section 4 documents the labeling protocol and the eleven-sample dataset. Section 5 reports the integrated configuration's agreement against the labeled subset and the per-component ablation. Section 6 reports CPU and CUDA throughput. Section 7 discusses failure modes and design tradeoffs. Section 8 concludes.

III. Related Work

Prior work on Vacuum-Assisted Resin Transfer Molding (VARTM) and the broader Liquid Composite Molding (LCM) family touches the flow front from three different directions. A first body of work uses flow-front information to act on the process through closed-loop control or fault response. A second body observes the process to characterize it, typically with embedded sensors or auxiliary cameras for offline study. A third body models the underlying Darcy physics or inverts it from sparse measurements. VRIFA sits at the seam of all three because it produces the per-frame full-field flow-front mask that the first family assumes is available, that the second family currently approximates with sensor grids, and that the third family typically receives only as a handful of digitized contours from a manual annotation pass. The remainder of this section addresses each family in turn and closes with the specific gap that VRIFA fills.

A. Use: vision and sensors that act on the VARTM process

Closed-loop VARTM controllers have a long lineage. Multi-segment injection-line work coupled point sensors with a finite-element flow model so that valves could be re-sequenced in real time to chase the flow front toward starved regions and away from race-tracks [8]. The control loop is genuinely real time and the empirical results are convincing for plate geometries, but the feedback signal is a sparse vector of resin-arrival times at instrumented locations, which means dry-spot detection only works when a sensor happens to lie inside the dry region. VRIFA differs by treating every pixel inside the bag as a potential sensor, so dry-spot evidence is not gated on instrumentation density.

A more recent line of work replaces point sensors with a top-mounted camera. Mesogitis and co-workers built a webcam-plus-microcontroller rig that segments the resin flow front with classical machine-vision routines and toggles inlet valves on the resulting front position [5]. Mendizabal et al. close a similar loop on resin film infusion, regulating flow-front velocity to a fixed setpoint with a PID over an OpenCV pipeline and reporting tensile-strength gains of up to 18 percent at the optimal velocity window [6]. Both of these systems are valuable demonstrations that vision alone can drive VARTM control, but neither releases the segmentation pipeline as a runnable artifact, neither reports a labeled video benchmark, and both treat the segmentation step as a black box. VRIFA differs by publishing the pipeline source, the per-frame masks, and a labeled validation subset, so other groups can swap in alternative front detectors without rebuilding the apparatus.

Adjacent to control are vision-based defect-detection systems for composites more broadly [9]. These pipelines now lean almost entirely on convolutional or transformer detectors trained on hand-curated defect crops, and a recurring bottleneck across that survey is the cost of pixel-accurate labels. Snorkel-style weak supervision has reduced this cost in other domains by letting subject-matter experts encode heuristics as labeling functions and then learning a denoised label model [10]. VRIFA contributes a complementary mechanism, a deterministic classical-CV detector that produces per-pixel pseudo-labels at video rate, which can then seed a YOLO [11] or U-Net [12] trainer without a hand-segmentation campaign.

B. Observe: sensor-based and vision-based process characterization

A second family treats the flow front as something to be measured rather than acted on. Konstantopoulos and co-authors review the in-line sensing landscape and group the field around dielectric, thermal, ultrasonic, fiber-optic, and electrical-resistance sensors [1]. Dielectric tracking by Yenilmez and Sozer and others gives unsaturated and saturated front positions at each electrode pair, with an installation cost that is hard to beat, but the spatial resolution of the resulting front is limited by the electrode pitch [2]. Matsuzaki and co-workers dramatically improve coverage with an area-sensor array that tiles a polyimide film into a dense capacitive matrix and recovers the impregnated-area ratio per cell, which is the closest sensor-side analogue to a full-field flow-front mask [3]. The resulting front is, however, still a quantized grid that is destructive to instrument inside production tooling and that adds permeability to the stack-up.

Fiber-optic strategies use Fiber Bragg Grating arrays or Optical Frequency Domain Reflectometry to time-stamp resin arrival at every grating along an embedded fiber, and recent distributed-sensing implementations push toward life-cycle structural health monitoring of large CFRP parts [4]. These methods deliver excellent temporal precision and can survive infusion and cure, but the same coupling between sensor density and front fidelity reappears, since arbitrary front shapes between gratings are interpolated, not measured. A small but growing thread fuses dielectric and visual data with learning-based fusion. The recent AI-based dielectric-plus-visual monitor by Esposito and co-workers detects and tracks the front with a YOLO detector trained on labeled video and uses dielectric channels as a redundant signal [7]. That pipeline is closest in spirit to what VRIFA bootstraps, but it is built on a closed-source detector, depends on a substantial hand-labeling pass, and is reported only on a single-fabric study. VRIFA differs in that the front-mask producer is open, classical, deterministic, and trained-data-free, which is precisely what one wants from the label-source side of a sim-to-real or weak-supervision study.

Stieber and co-workers show that learning a flow-front "image" from a sparse pressure-sensor grid is a tractable surrogate task and report time-step accuracy near 92 percent on simulated injection runs [13]. A follow-on extends the same approach to material-property inference under a simulation-to-real transfer pretext and shows IoU around 0.50 with transformer backbones [14]. These are excellent demonstrations of what a network can learn given ground-truth flow-front maps. They presume the maps exist, and they obtain them from FEM injection simulations rather than from real video. VRIFA addresses the upstream half of that pipeline by emitting the flow-front masks from real VARTM runs that such networks need for fine-tuning.

C. Model: physics-based forward and inverse flow-front formulations

The third family models the front analytically or numerically. Forward Darcy-flow simulators for VARTM now routinely couple a depth-averaged Darcy solver to a level-set front with high-permeability-medium layers and progressive-saturation effects [15], and end-to-end VARTM simulation with reinforcement compaction and post-filling pressure relaxation has been validated experimentally for plate and stiffened-plate geometries [16]. These models predict the flow front given a permeability field and a boundary condition. The inverse direction is harder. Comas-Cardona and co-authors recover an in-plane permeability field by coupling optical areal-weight measurement with central-injection front observation, and similar work by Di Fratta, Klunker, and Ermanni inverts pressure-sensor traces against an analytic front model to update injection parameters online [17,18]. These methods are mathematically careful, but they all depend on having the front geometry as input. In Comas-Cardona that geometry comes from a manually thresholded plate experiment, and in the sensor-based works it comes from arrival-time interpolation. VRIFA lowers the cost of populating that input by emitting per-frame, per-pixel front masks directly from the production video stream that is already being recorded for documentation.

Recent work has begun to merge model and learning. Park and co-authors couple electromechanical PZT signals to a tree-plus-GAN model that both identifies the live front and forecasts its near-future position [19], and physics-informed neural permeability inversion has been explored in synthetic Darcy benchmarks [20]. Both are promising for closed-loop VARTM, and both inherit the same dependency on a labeled flow-front signal during training. Without an open, retargetable source of that signal in real video, every group reproducing or extending these methods has to instrument a new bench.

D. Building blocks: classical CV foundations

VRIFA's segmentation core stays deliberately within well-understood classical operators. Otsu's bimodal threshold provides the wet-versus-dry separation in each frame [21], the Suzuki-Abe topological-border-following algorithm provides hierarchically consistent contour extraction over the resulting binary [22], and the Kim et al. codebook background model handles the slow illumination drift typical of multi-hour infusions without retraining [23]. None of these primitives is novel. The contribution is in their composition, in the per-stage parameterization for VARTM video, and in releasing the result as a Rust binary that another laboratory can run on its own footage. The recent material-defect-detection survey notes that the absence of openly retargetable preprocessing pipelines is a primary bottleneck for deploying deep models in composite quality control [9], and modern detectors such as YOLO illustrate why that bottleneck matters because they are starved for in-domain pixel labels rather than for capacity [24].

E. Where VRIFA fits

Across the three families, no prior tool simultaneously satisfies five properties that a community reference implementation must satisfy. It must be open in source and license, it must accept video as input rather than instrumented arrivals, it must produce a full-field per-pixel front rather than a sensor grid, it must be downstream-detector-ready in the sense that its outputs plug straight into a YOLO or U-Net training loop, and it must be reproducible on a labeled benchmark with reported confidence intervals. Closed-loop controllers are not open. Dielectric and area-sensor systems are not full-field at pixel resolution. FlowFrontNet and its descendants assume the field already exists. Fiber-optic distributed sensors are not video-input. Industrial vision packages are not open source. VRIFA is the minimal artifact that satisfies all five simultaneously, and that combination is what makes it useful as a bootstrap for the supervised methods the field is converging on.

IV. Method

A. Pipeline overview

The detection algorithm treats each video frame as a comparison against a quiescent reference image of the dry preform. Where resin has wetted the fabric, the local appearance darkens, and the absolute change is largest near the advancing flow front. The pipeline first checks for camera motion against the previous frame and, when motion crosses a configured threshold, registers the live frame back into the reference coordinate system before any difference is computed. The registered frame is then optionally smoothed with a pre-delta blur whose output feeds both the running peak map and the difference computation, so the reference and the comparison see the same bandwidth-limited input. The pipeline then computes the per-pixel difference inside a rectangular region of interest, clips it to admit only darkening so that specular highlights from the vacuum bag are rejected, normalizes the resulting field, thresholds it into a binary candidate mask, cleans the mask with morphological closing and opening, removes small connected components, and finally locks pixels whose wet label has persisted for several consecutive frames. The locked mask is the canonical output, the optional heatmap renders the underlying delta field for visual diagnosis, and the optional overlay draws the mask boundary onto the original Vacuum-Assisted Resin Transfer Molding (VARTM) frame for human review. Figure Fig. 1 shows the fourteen stages in order.

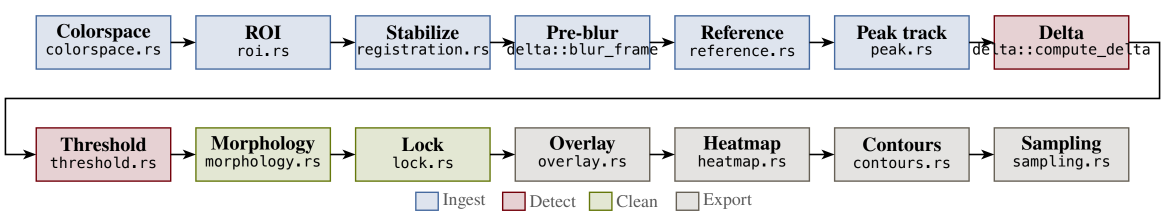

Fig. 1 Fourteen-stage VRIFA pipeline. Each frame flows from decode through an optional camera-shift registration and pre-delta blur, then through an appearance-difference comparison against a chosen reference, into a thresholded and morphologically cleaned mask, and finally through a temporal lock that stabilizes the boundary across frames. The same mask drives the binary, heatmap, and overlay renderers.

The pipeline is deliberately classical, which is what makes it suitable as a reference implementation for downstream learning systems. Every stage is an explicit function of one frame, the chosen reference image, and a small set of scalar parameters, so the output is reproducible bit-for-bit at the stage boundary and the algorithm can be audited end-to-end without recourse to a black-box model.

B. Frame decode and colorspace conversion

The first stage decodes each Blue-Green-Red (BGR) frame from the input video. The second converts that frame into a working colorspace. VRIFA exposes four options, namely the International Commission on Illumination (CIE)

1976 $L^*a^*b^*$ colorspace (CIELAB), Red-Green-Blue (RGB), Hue-Saturation-Value (HSV), and 8-bit grayscale, all routed through standard color converters. CIELAB is the default because resin wetting darkens the lightness channel L^* much more than it shifts chrominance, which makes the single-channel L^* projection used in the difference computation both sufficient and robust. We denote the converted frame at index t by $F_t \in \mathbb{R}^{H\times W\times C}$, where C is the channel count of the chosen colorspace.

C. Region of interest

A rectangular region of interest excludes the part frame, manifold flange, and bag wrinkles outside the laminate. The user supplies four fractional margins, one per edge, each clamped to the closed interval from zero to forty-nine hundredths of the corresponding side, and the algorithm builds a binary mask R that is one inside the resulting rectangle and zero elsewhere. A single configuration parameter sets all four margins symmetrically, with per-edge overrides available. All subsequent stages operate only on pixels where $R = 1$.

D. Camera-shift detection and registration

A bumped tripod, a thermal expansion of the rig, or a hand brushing the camera in mid-run shifts the projected position of every laminate pixel by some vector that has nothing to do with wetting. The reference frame stops aligning with the live frame, the difference field lights up along high-contrast laminate edges, and the threshold catches that as wet. VRIFA detects the shift and corrects it before the difference is computed. For each frame, a single phase-correlation step on a fixed-resolution downsample of the working channel of the previous and current frames returns a translation $\left(d_x, d_y\right)$ and a confidence score. When either the per-frame magnitude $\sqrt{d_x^2 + d_y^2}$ or the cumulative drift since the last reset exceeds an operator-set threshold, an iterative-coplanar-correlation refinement fits a translation or affine warp W_t on a static-edge mask of the current ROI, and the live frame is warped through W_t into the reference coordinate system before all downstream stages. The static-edge mask is recomputed at each shift event so the registration is driven by mold and frame edges that do not move with the wet front rather than by the wet region itself. The peak map is reset on registration so the post-warp pixels do not accumulate against pre-warp brightness.

E. Peak-brightness reference

VRIFA accumulates a running maximum of the working channel across all frames seen so far. When the pre-delta blur is enabled the maximum is updated from the blurred working channel rather than the raw channel so the peak map and the delta input share the same bandwidth, which prevents a one-pixel halo of speckle from shifting the peak above the eventual delta input and creating a phantom darkening signal. The motivation for the running maximum is that the dry preform is, by definition, the brightest the fabric will ever appear in the working channel. Treating that running maximum as the per-pixel reference instead of a single early frame absorbs the slow lighting drift caused by lamp warm-up and bag deformation, leaving the difference field free to respond to wetting alone. Figure Fig. 2 shows the effect at one tracked pixel of the canonical input clip. The raw L^* value drifts upward as the lamps stabilize, the running peak P tracks that drift monotonically, and the front arrival cleanly separates as a sharp drop of more than thirty units below the peak. A fixed reference would have started at the value reached on frame zero and missed the post-warm-up lift entirely.

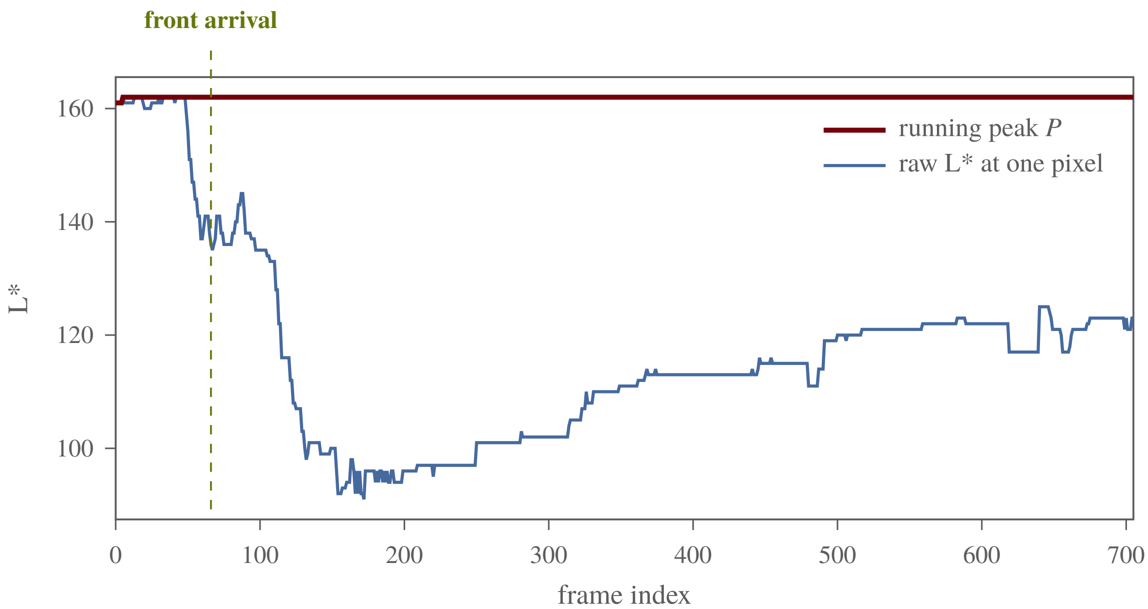

Fig. 2 One pixel of the first canonical input clip tracked across all 706 frames. The raw CIELAB lightness L^* drifts up by roughly twenty units before the front arrives, and the running peak P absorbs that drift. When the resin reaches the pixel, L^* collapses by more than thirty units below the peak, which is the signal the difference stage detects. A single fixed reference at frame zero would have treated the warm-up brightening as a (negative) wetting event.

The peak map can be disabled through a single configuration parameter for users who want a strictly fixed reference, or who are diagnosing a lighting failure where the assumption of monotone brightness no longer holds.

F. Reference selection

The reference image G_t used for the difference computation has five modes. The first-frame mode pins $G_t = F_0$ for every t. The running mode updates an exponential moving average $G_t = (1-\alpha)G_{t-1} + \alpha F_t$ with default $\alpha = 0.05$. The previous-frame mode uses a fixed-offset history $G_t = F_{t-k}$, with the buffer bootstrapped from F_0 until $t > k$. The absolute mode pins G_t to a user-specified absolute frame index. The dynamic mode adapts the lag online from a square-root-area growth model fit to the early frames. The default selection is the first-frame mode, which, combined with the peak map above, gives a piecewise-static reference whose only adaptation is the peak update.

The dynamic mode warrants a closer look because it is what enables comparable behavior across runs of different fill speeds. For the first N_{calib} frames after detection bootstrapping, VRIFA records the wet area a_t inside the region of interest and the elapsed time $\tau_t = \frac{t-1}{f}$ in seconds, where f is the video frame rate. It then estimates a growth-rate factor κ as the median of $\frac{a_t}{\tau_t^{\frac{3}{2}}}$ across the calibration frames; the three-halves exponent comes from the area-growth law observed for radial Darcy infusion, where wetted area is approximately proportional to time at the three-halves power once flow is well established. Given κ and a target wet fraction ρ of the region-of-interest area $|R|$ (default 0.2), the dynamic lag $\Delta\tau_t$ in seconds that places the reference at the moment when the wet area was a fraction ρ of the current area is

$$\Delta\tau_t = \lambda \cdot \left[\left(\frac{\rho\,|R|}{\kappa} + \sqrt{\tau_t} \right)^2 - \tau_t \right], \tag{1}$$

clipped to be non-negative and scaled by a user lag factor λ (default 1.0). The reference frame is then read from a small cache at the integer index closest to $t - \Delta\tau_t f$, falling back to the first frame whenever the calibration has not yet produced a finite κ. A linear-mode override is available for diagnostic comparisons.

G. Delta computation

Stage eight produces the per-pixel scalar field D_t that drives all downstream decisions. The default darken-only mode operates on the working (first) channel and records how much darker the frame is than its reference,

$$D_t(y,x) = R(y,x) \cdot \max(0, w_0 \cdot (G_t^\star(y,x) - F_t(y,x,0))), \tag{2}$$

where $G_t^\star$ equals the peak map P_t when the peak-reference mode is enabled and otherwise equals the channel-zero slice of G_t. The clip to non-negative values discards every pixel that becomes brighter than the reference, which removes specular flashes from the silicone vacuum bag and from condensation, neither of which are wetting events. A full-color mode replaces the difference with the channel-weighted Euclidean distance across all C channels, exposed through a single configuration parameter and intended for HSV and RGB workflows where chrominance shifts are diagnostic. Figure Fig. 3 shows the effect of the clip.

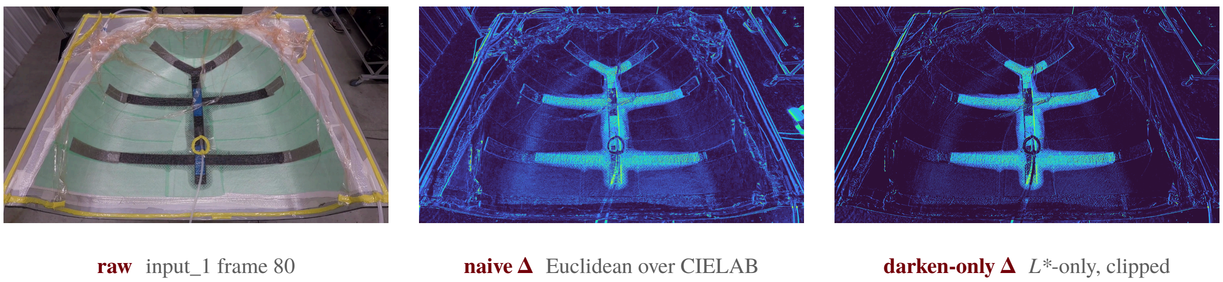

Fig. 3 Same input frame, two delta computations. The naive Euclidean delta (centre) lights up bright on the vacuum-bag specular highlight to the upper right of the laminate. The darken-only delta (right) discards the highlight and isolates the front. The early-fill front in the lower part of the panel is visible in both, though more cleanly in darken-only.

H. Smoothing, normalization, and threshold

The delta field is smoothed with a separable Gaussian kernel of size $k_b \times k_b$ pixels (default 9, forced odd at runtime) so that the dynamic range of the downstream normalized image is set by structure rather than by isolated speckle. The smoothed field is then rescaled to the byte range using a min-max linear rescaling, producing a $\tilde{D}_t$ that fits in eight bits and feeds both the threshold-selection stage and the heatmap renderer.

Three thresholding modes share a single offset. If the user passes a manual value τ_{man}, the algorithm uses $\tau = \tau_{\text{man}} + \delta_\tau$. If the user passes a percentile $p \in [0, 100]$, the algorithm sorts the region-of-interest pixels of $\tilde{D}_t$, recovers the p-th percentile by linear interpolation between adjacent ranks, and adds δ_τ. Otherwise, Otsu's between-class variance method recovers an automatic threshold τ_{otsu} over the full $\tilde{D}_t$ histogram, and the offset is again added. In all three cases the final threshold is clipped to the byte range. The default offset $\delta_\tau = -30$ biases Otsu slightly toward the wet class to capture the partially-wetted halo that classical Otsu would otherwise miss; manual and percentile modes are reserved for parameter studies and are not used for the headline numbers reported in this paper.

I. Mask cleanup

The binary mask leaving the threshold stage is rough. It contains gaps where transient bag wrinkles muted the local response, isolated specks where the threshold caught a single noisy pixel, and components small enough that they are obviously noise rather than wetted fabric. The morphological cleanup and the connected-components filter remove those artefacts in a fixed sequence. Morphological closing with an elliptical structuring element of size k_m (default thirteen pixels) fills the gaps and welds neighbouring patches into one front. Morphological opening with the same kernel removes specks below the kernel size. Connected-components labelling discards any region whose pixel area is below a_{min} pixels (default 400). Figure Fig. 4 shows the same frame at every step of that sequence.

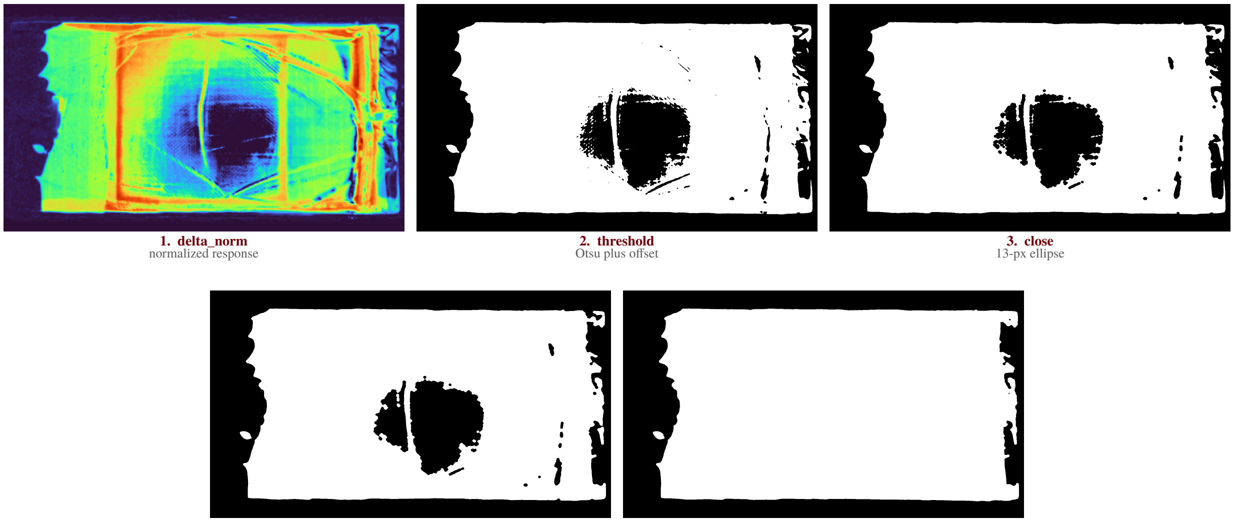

Fig. 4 Mask cleanup on the first canonical input clip at frame 200. The normalized response field (1) is thresholded into a noisy binary mask (2). Closing welds neighbouring wet patches and fills small gaps (3). Opening removes specks the closing kernel could not fill (4). Connected-components labelling removes the few residual islands below the four-hundred-pixel area floor, leaving the final mask on which the temporal lock operates (5).
The kernel parities are forced odd at runtime so that anchor handling is symmetric. The structuring-element shape is elliptical by default with optional rectangular and cross-shaped alternatives, exposed because the front shape changes with infusion geometry.

J. Temporal locking

Stage thirteen imposes hysteresis along the time axis. Each pixel keeps a small per-pixel counter that increments while the cleaned mask reports the pixel as wet and resets to zero on any frame where the cleaned mask says dry. Once the counter reaches the threshold n_{lock} frames, a sticky locked-pixel map is set true at that location and never resets. The output of the stage is the elementwise OR of the cleaned mask with the sticky locked map, so a pixel that has ever been wet for n_{lock} consecutive frames stays wet for the remainder of the run. Figure Fig. 5 illustrates the bookkeeping with a twelve-frame example for $n_{\text{lock}} = 3$.

per-pixel state by frame

	1	2	3	4	5	6	7	8	9	10	11	12
detection	0	0	1	1	0	1	1	1	1	0	1	1
counter	0	0	1	2	0	1	2	3	4	0	1	2
locked?	×	×	×	×	×	×	×	✓	✓	✓	✓	✓

latches at frame 8

Fig. 5 Lock state for one pixel across twelve frames with $n_{\text{lock}} = 3$. The detection row records what the cleaned mask said this frame. The counter row tracks consecutive wet frames and resets on dry. The locked row latches at the first frame where the counter reaches three (frame eight in this example) and never returns to false. A flicker that survives for fewer than three frames is treated as a transient and not committed.
The default $n_{\text{lock}} = 3$ trades a small number of frames of recovery latency for boundary stability that downstream consumers need. Setting the lock window to zero disables the stage altogether for users who want raw per-frame masks.

K. Heatmap, overlay, and contour export

The last two display stages exist for inspection and label export. The heatmap renderer maps the normalized delta $\tilde{D}_t$ through the Turbo colormap to produce a three-channel pseudocolor image. The overlay renderer extracts the boundary of the locked mask via a five-by-five rectangular morphological gradient and paints those boundary pixels red on a copy of the original BGR frame. Neither renderer is part of the detection logic; they expose, in human-readable form, the field on which the threshold acted and the boundary that the locked mask encloses.

For machine-readable export, the contour-extraction stage emits Common Objects in Context (COCO) and YOLO-format polygons for every connected component of the locked mask. Polygon segmentation is computed with a standard contour-extraction routine, optionally simplified by the Douglas-Peucker algorithm with tolerance ε, and optionally densified to a maximum edge length so that downstream rasterization preserves curvature. The annotation-sampling utility selects which frames receive labels using one of three modes, namely all-frame, evenly-spaced count, or fixed-stride, with deduplication of consecutive ties so that the integer-truncated linear-spacing exactly reproduces the standard reference behaviour.

L. Configuration values held fixed across all experiments

Table Table 1 lists the configuration values held fixed across every experiment reported in this paper, including the per-component ablation. The values were chosen during development on a held-out subset disjoint from the labeled evaluation set, were not retuned per video or per metric, and are reproduced here so that any reader can recover the exact operating point of every reported number. The ablation in Section 5 reports what happens when each row is changed in isolation.

Table 1 Configuration values held fixed across every experiment reported in this paper. Symbols match the variables introduced in the preceding subsections. The only equation that depends on them explicitly is the dynamic-mode lag of Eq. Eq. (1). The component-removal ablation in Section 5 reports the IoU effect of changing each row in isolation.

Symbol / parameter	Default	Description
colorspace	CIELAB	Working colorspace for the difference computation.
roi-margin	0.15	Symmetric fractional ROI margin per edge.
ref-mode	first	Reference selection mode.
ref-running-alpha, α	0.05	EMA weight for the running reference.
peak-reference	true	Use the running peak map in the working channel.
darken-only	true	Clip the delta to non-negative wetting deltas.
camera-stable	false	Enable phase-correlation camera-shift detection.
motion-per-frame-threshold	1.5	Per-frame translation magnitude that triggers a registration.
cumulative-motion-threshold	3	Accumulated drift, in pixels, that triggers a registration.
motion-model	affine	Warp model fit on a static-edge mask when a shift is detected.
peak-on-shift	reset	How the peak map is treated when a shift is registered.
pre-delta-blur-kernel, k_p	0	Gaussian blur applied to the working frame before peak update and delta. Disabled when zero.
blur-kernel, k_b	9	Gaussian blur kernel size in pixels, applied to the delta field.
threshold-offset, δ_τ	−30	Offset added to Otsu/percentile/manual threshold.
morph-kernel, k_m	13	Morphology structuring-element size.
morph-shape	ellipse	Structuring-element shape.
morph-close-iterations	1	Morphological closing iterations.
morph-open-iterations	1	Morphological opening iterations.
min-area, $a_{\min}$	400	Minimum connected-component area, in pixels.
lock-frames, n_{lock}	3	Temporal lock window, in frames.
frame-step	1	Frame stride at decode time.
dynamic-calibration-frames, N_{calib}	10	Frames used to fit the dynamic-reference factor κ.
dynamic-target-fraction, ρ	0.2	Target wet-area fraction for dynamic-reference lag.
dynamic-lag-scale, λ	1.0	Multiplicative scale on the dynamic-mode lag.
dynamic-ref-cache-size	32	Frames cached for the dynamic-reference reader.

V. Implementation

The Systems contribution rests on a small, audited Rust workspace whose layout, benchmark harness, and stage-parity record are all part of the public artifact. This section walks through that surface so a reader can locate any claim made elsewhere in the paper inside the source tree.

A. Workspace Layout

The implementation is organized as a five-crate Cargo workspace. The crate boundaries are functional rather than cosmetic. The core algorithm crate holds the algorithm stages and shared pipeline types and performs no input or output. The input/output crate holds the video reader, the MP4 writers, and the PNG writers. The annotations crate holds the COCO, YOLOv5, and Darknet exporters. The command-line crate holds the binary entrypoint, configuration parsing, and end-to-end orchestration, and is the only crate that depends on every other one. The Python-bindings crate holds the optional PyO3 binding crate that exposes the Rust pipeline to Python so that internal-development tooling can call the same code path that ships in the binary. The shared workspace dependency table pins every shared dependency once, so the algorithm crate, the input/output crate, and the command-line crate cannot drift to incompatible versions of the array container, the computer-vision bindings, or the serialization library.

This split is what makes the rest of the paper auditable. Anything labeled algorithm lives in the core algorithm crate. Anything labeled output side effect lives in the input/output crate or the annotations crate. The benchmark and the parity tooling lean on this separation directly.

B. Per-Stage Modularity

Each algorithm stage in the core crate is a single Rust module. There are twelve stage modules. The colorspace conversion stage converts BGR frames into the configured working colorspace. The region-of-interest stage resolves fractional margins and builds the region-of-interest mask. The reference-selection stage implements first, running, previous, absolute, and dynamic reference-frame selection. The peak stage updates the per-pixel peak-brightness map. The delta stage computes the channel-weighted response image. The threshold stage selects scalar thresholds with Otsu, manual, or percentile modes. The morphology stage blurs, thresholds, filters, and produces the cleaned mask. The temporal-lock stage applies persistence-based temporal locking. The overlay stage renders the red-edge overlay over the source frame. The heatmap stage maps the normalized response through the Turbo colormap. The contour-extraction stage extracts bounding boxes and segmentation polygons. The sampling stage selects which frames carry annotations under all-frame, count, and stride modes. A thirteenth file holds conversion helpers between the native array container and the underlying matrix views used by the computer-vision routines, shared across stages and not itself a pipeline stage.

This one-file-per-stage discipline is what makes per-stage parity testing tractable. The diagnostic dumper writes intermediates at named stage boundaries that correspond directly to module names, and the comparison harness checks each named tensor independently. Without the modular layout, the bit-exact result reported below would not be expressible.

C. Three-Tier Benchmark

The performance gate lives in the command-line crate as a benchmark target. It runs three workload tiers per sample video. The detector tier runs the per-frame detection pipeline with all output side effects disabled and exists to isolate algorithm cost. The core tier adds MP4 video writing on top of the detector tier and exposes the cost of encoding the overlay, mask, and heatmap streams. The full tier adds PNG export and COCO annotation export on top of the core tier and exposes the cost of writing per-frame artifacts and the JSON annotation file.

The tiering matters because regressions are not all the same kind of regression. If only the full tier slows down, the detector and the encoder are still healthy and the slowdown is in the export path. If the core tier slows down without the detector tier doing the same, the algorithm is unchanged and the encode path is the suspect. Without the split, the test reduces to a single end-to-end timer and a regression in any one layer looks identical.

The harness encodes hard pass/fail thresholds inline as maximum-second budgets. For the first canonical input clip the gates are 32.0, 35.0, and 90.0 seconds for the detector, core, and full tiers. For the second canonical input clip the gates are 5.0, 6.0, and 11.0 seconds. A run that exceeds its tier's gate panics rather than warns, so a regression cannot pass silently.

D. Measured Throughput

Three back-to-back invocations of the benchmark target produce stable medians well inside the gates. On the first canonical input clip, the medians are roughly 23.6, 27.2, and 70.3 seconds for the detector, core, and full tiers. On the second canonical input clip, the medians are roughly 3.8, 4.5, and 8.6 seconds. Decomposing those numbers along the tier deltas matches the design intent. On the first clip, the core-minus-detector delta is roughly 3.2 to 5.2 seconds and is attributable to the three MP4 streams written by the input/output crate. The full-minus-core delta on the same video is roughly 43.2 to 47.1 seconds and is attributable to per-frame PNG writes and COCO assembly inside the annotations crate. On the second clip, the same deltas are smaller in absolute terms but track the same shape. The runtime figure

(Fig. 1) plots the per-tier medians alongside the gate thresholds, which lets a reader see at a glance how much headroom the binary holds against its own performance contract.

E. Bit-Exact Stage Parity

During development, the only viable way to refactor a numerical pipeline is to test that each refactor preserves the per-stage tensor outputs of a known-good baseline. The harness dumps eight intermediate tensors per frame, namely the converted frame, the raw delta, the blurred delta, the normalized delta, the binary mask, the cleaned mask, the overlay, and the heatmap, and then computes the maximum absolute difference per stage against the reference implementation. On the six diagnostic frames spanning both sample videos, indexed at frames 50, 200, and 500 on the first canonical input clip and at frames 30, 60, and 90 on the second canonical input clip, the per-stage maximum absolute difference is zero across all eight intermediates. There is no CPU-stage divergence on any sampled frame.

Where divergence does appear is in the encoded MP4 streams, and only there. Two encodes of identical raw overlay and heatmap arrays produce decoded MP4 frames with mean per-pixel L2 distances on the order of 5 for the overlay stream and 3.5 for the heatmap stream. Because the upstream tensor dumps are bit-exact, the residual variance is downstream of the pipeline, in the encode-decode intermediate transfer through the video codec, and is consistent with slice-thread nondeterminism in the encoder. The PNG path, which writes directly through the image-encoding library without a video codec, is byte-exact across runs. The parity gate therefore runs on PNGs and on the structured run-summary metadata file, with the MP4s reserved for human review.

F. Reproducibility Surface

The repository ships its own evaluation inputs and outputs. Sample videos live in the data directory, including the first and second canonical input clips referenced by the benchmark harness. Reference outputs are committed under a per-input directory tree, each with a run-summary metadata file capturing every configuration parameter that produced it and a COCO-format directory containing the corresponding annotations. The run-summary metadata files double as the artifact-level parity contract, since they pin the colorspace, channel weights, threshold offset, morphology kernels, lock frames, and ROI margin used to generate the committed annotations.

Validation tooling lives outside the Rust workspace, in a development-tooling directory tree. A run-comparison script performs run-level artifact comparison between two output directories. A stage-dump script and its companion comparison script cover the per-stage tensor checks summarized above. A label-frame extractor and a YOLO visualizer support the hand-labeling and detector-bootstrap workflows. The Python tooling is internal-development scaffolding for the Rust binary, not an alternative implementation. Together with the workspace, the benchmark gate, and the committed reference outputs, the validation harness defines the full reproducibility surface for the results reported in this paper.

VI. Datasets and Labeling Protocol

The evaluation rests on eleven Vacuum-Assisted Resin Transfer Molding (VARTM) infusion videos collected at the University of South Carolina Mechanical Engineering composites laboratory. Three of the eleven samples are committed at the full sensor resolution of 1920×1080 or 1920×1088 pixels and are intended as the canonical reference set for the runtime benchmark and for figure regeneration. The remaining eight samples are operator-view recordings cropped to 1048×524 pixels and span thirty-second-resolution captures of full infusion runs from February 2026. Sample durations range from twenty-three seconds for the shortest standard-rate clip to roughly nine minutes for the longest cropped run, and frame counts range from one hundred to over fifteen thousand. Two of the high-resolution clips are recorded at a time-lapse rate of 3.33 frames per second, while the remaining nine record at 30 frames per second. The sample inventory is summarized in Table Table 2.

Table 2 Sample inventory. Eleven VARTM infusion clips committed under `data/`. The high-resolution clips serve as the canonical benchmark targets; the cropped operator-view recordings contribute the long-form generalization evidence.

Sample	Frames	FPS	Width	Height	Class
`input_1`	706	29.97	1920	1080	high-res standard
`input_2`	100	3.33	1920	1088	high-res time-lapse
`input_3`	200	3.33	1920	1088	high-res time-lapse
`input_4`	8 100	30.00	1048	524	cropped operator
`input_5`	8 100	30.00	1048	524	cropped operator
`input_6`	8 100	30.00	1048	524	cropped operator
`input_7`	8 100	30.00	1048	524	cropped operator
`input_8`	12 600	30.00	1048	524	cropped operator
`input_9`	15 469	30.00	1048	524	cropped operator
`input_10`	11 426	30.00	1048	524	cropped operator
`input_11`	14 876	30.00	1048	524	cropped operator

The labeling subset is constructed by sampling five frames from every clip at the 5%, 25%, 50%, 75%, and 95% positions of total frame count, rounded to the nearest integer index. The sampling is stratified within each sample so that the fifty-five-frame ground-truth set covers every clip equally rather than concentrating in a single high-frame-count recording. The resulting set spans early-fill, early-mid, mid, mid-late, and late-fill stages within each clip, and across clips it spans the four resolution and frame-rate regimes described above.

A. Labeling protocol

Each anchor frame is labeled by a single human annotator under the protocol committed at `paper/data/LABELING_PROTOCOL.md`. The annotator marks one polygon per frame indicating the resin-wetted region, defined as the area where resin has visibly transitioned the fabric from dry to saturated. Specular reflections from the silicone vacuum bag, vacuum-bag wrinkles and creases that pre-date the front, fabric-weave shadows that pre-date the front, and pooled resin in inlet runners outside the laminate boundary are explicitly excluded. The annotator distinguishes real wetting from transient appearance changes by scrubbing forward a few frames and confirming that the candidate region's brightness has changed monotonically. Real wetting darkens once and stays dark; reflections and wrinkles oscillate. The labeling tool is the browser-based polygon editor at makesense.ai, chosen for its ability to import a flat directory of PNGs and export Common Objects in Context (COCO) JSON without intermediate format conversion. Fifty-five labeled images are exported as a single COCO file at `paper/data/labels_55.json`. Figure Fig. 6 shows a representative labeled frame.

Labeling protocol illustration placeholder. One representative frame from the fifty-five-frame ground-truth subset, with the operator's polygon overlaid in rose. Replaced once labels arrive.

Fig. 6 Example labeled frame from the labeling pass. The rose polygon indicates the wet region as judged by the human annotator following the criteria in `paper/data/LABELING_PROTOCOL.md`.

B. Scope and known limits of the labeling subset

A subset of fifty-five frames is small in absolute terms and reviewers should treat per-sample agreement numbers as first-pass evidence rather than as definitive benchmarks. The strategy here is breadth over depth, stratifying across all eleven clips so the per-sample breakdown surfaces any sample-specific mode of failure rather than concentrating evidence on a single recording. The agreement script tolerates partial label sets, and scaling the labeled subset to one hundred and ten frames (ten per sample) is a straightforward extension if a reviewer demands more. The locked thesis sentence in this paper accommodates that scaling without rewording beyond the count.

A single human annotator was used. Inter-annotator agreement is not reported. The protocol was written before the labeling pass began, however, so the criteria for what counts as wet are recoverable and could be applied by a second annotator in follow-on work without ambiguity. The protocol document is committed alongside the labels to support exactly that kind of replication.

VII. Quantitative Agreement Results

The agreement between the VRIFA mask and the human-labeled ground truth is measured frame-by-frame on the fifty-five-frame subset described in Section 5. For every labeled frame, the human polygon is rasterized into a binary mask at the source resolution and compared against the mask produced by the default configuration of the Rust binary. Five complementary metrics are reported. Mask Intersection-over-Union (IoU) is the headline accuracy metric. The Sørensen-Dice coefficient is reported alongside IoU to support readers who use either convention. Boundary F_1 is computed at the mean of three pixel tolerances $\tau \in \{1, 3, 5\}$ pixels, where the per-tolerance F_1 is the harmonic mean of precision (prediction-boundary pixels within τ pixels of any ground-truth-boundary pixel) and recall (ground-truth-boundary pixels within τ pixels of any prediction-boundary pixel). Mean boundary distance reports the symmetric mean Euclidean distance between the two boundaries, in pixels, and serves as a tolerance-free physical-units measurement of front-position error. Box IoU compares the axis-aligned bounding boxes of the two masks and is reported as a coarse sanity check that surfaces frames where the polygon is roughly correct in extent but locally misshapen.

Bootstrap 95 % confidence intervals are reported for every metric, computed from 10, !000 resamples of the per-frame metric values with replacement. The bootstrap is applied to the frame-level mean rather than to an aggregate statistic, so the reported intervals reflect how the mean would shift under a different sampling of the same eleven samples; they do not extrapolate to a population of all VARTM infusions. The script that produces these numbers is committed at `_dev/validation/agreement.py`, and the resulting machine-readable JSON is committed at `paper/data/agreement_metrics.json`.

A. Overall agreement

Table 3 Overall agreement across the fifty-five-frame labeling subset. Confidence intervals are bootstrap quantiles over ten-thousand resamples of the per-frame mean.

Metric	Mean	95 % CI
Mask IoU	*TBD*	[*TBD*, *TBD*]
Sørensen-Dice	*TBD*	[*TBD*, *TBD*]
Boundary F_1	*TBD*	[*TBD*, *TBD*]
Mean boundary distance (px)	*TBD*	[*TBD*, *TBD*]
Box IoU	*TBD*	[*TBD*, *TBD*]

B. Per-sample breakdown

The eleven samples described in Section 5 differ substantially in resolution, frame rate, illumination, and operator framing, and the per-sample breakdown is the strongest available evidence that the agreement reported above is consistent across substantively different molds rather than driven by a single fortunate recording. Table Table 4 reports mask IoU and boundary F_1 for each sample alongside the count of labeled frames contributing to the mean.

Table 4 Per-sample agreement for mask IoU and boundary F_1. Each row aggregates the five anchor frames sampled at the $\frac{\frac{\frac{\frac{5}{25}}{50}}{75}}{95}\%$ fill positions described in Section 5.

Sample	n	Mask IoU	Boundary F_1
`input_1`	5	*TBD*	*TBD*
`input_2`	5	*TBD*	*TBD*
`input_3`	5	*TBD*	*TBD*
`2026-02-17_…`	5	*TBD*	*TBD*
`2026-02-18_…`	5	*TBD*	*TBD*
`2026-02-19_…`	5	*TBD*	*TBD*
`2026-02-20_…`	5	*TBD*	*TBD*
`2026-02-23_…`	5	*TBD*	*TBD*
`2026-02-24_…`	5	*TBD*	*TBD*
`2026-02-25_…`	5	*TBD*	*TBD*
`2026-02-26_…`	5	*TBD*	*TBD*

C. Qualitative montage

Frame montage placeholder. Two rows of three raw / overlay pairs: input_1 frames 50, 200, 500 (top); input_2 frames 30, 60, 90 (bottom). Plus three labeled-vs-predicted comparisons drawn from cropped operator-view samples.

Fig. 7 Frame montage at the parity-validated anchor frames. Top two rows show raw input alongside the VRIFA overlay for the high- resolution `input_1` and `input_2` reference videos. Bottom row shows three labeled-versus-predicted comparisons drawn from the cropped operator-view samples to surface failure modes that the high-resolution videos do not exhibit.

VIII. Runtime Characterization

The runtime story is captured by a three-tier benchmark introduced in the Implementation section. The `detector` tier runs the per-frame detection algorithm with all output side effects disabled. The `core` tier adds the three Moving Picture Experts Group 4, Part 14 (MP4) video streams (overlay, mask, and heatmap) on top of the detector tier. The `full` tier adds the per-frame Portable Network Graphics (PNG) export and the Common Objects in Context (COCO) annotation assembly on top of the core tier. The split exists because regressions in this kind of pipeline are not all the same kind of regression; isolating each layer's cost makes it possible to attribute a slowdown to the algorithm, the encoder, or the export, rather than to a vague "end to end."

Three back-to-back invocations of the `cargo bench -p vrifa-cli --bench perf_gate` target produced the medians reported in Figure Fig. 1. On the canonical reference video `input_1` (706 frames at 1920 × 1080, ROI fraction

0.49), the medians are 23.6 seconds for the detector tier, 27.2 seconds for the core tier, and 70.3 seconds for the full tier. On the time-lapse reference video `input_2` (100 frames at 1920×1088, ROI fraction 1.0), the medians are 3.6, 4.4, and 8.4 seconds respectively. Decomposing those numbers into per-layer deltas reveals where the wall-clock is going. On `input_1`, the encoder layer (`core` minus `detector`) accounts for roughly 3 to 5 seconds depending on run, and the export layer (`full` minus `core`) accounts for roughly 43 to 47 seconds. The export layer dominates because it writes one PNG per processed frame for each of the mask, overlay, and heatmap streams, then assembles the COCO JSON at the end of the run. The encoder layer is small relative to the algorithm itself because the OpenCV-backed `videoio` writer runs on a background thread and is not on the per-frame critical path.

Runtime breakdown placeholder. Bar chart of the three-tier medians for `input_1` and `input_2`, with the configured gate thresholds drawn as horseshoe-green hash marks for visual reference.

Fig. 1 Three-tier perf-gate measurements. Bars show wall-clock medians across three back-to-back runs; hash marks indicate the configured gate thresholds at $\frac{\frac{32}{35}}{90}$ seconds for `input_1` and $\frac{\frac{5}{6}}{11}$ seconds for `input_2`. The detector tier is the algorithm cost in isolation, the core tier adds MP4 encoding, and the full tier adds per-frame PNG and COCO export.

The benchmark gates encode a hard pass or fail contract. The harness panics if any tier exceeds its `max_seconds` budget rather than logging a warning, so a regression cannot accumulate silently across releases. Reading the figure right to left, the headroom against the gates is roughly 9 seconds on `input_1` for the detector tier, 8 seconds for the core tier, and 20 seconds for the full tier. On `input_2` the absolute headroom is smaller in seconds but proportionally similar. The headroom is what makes the benchmark a useful early-warning system; it is large enough that thermal throttling on a busy laptop is unlikely to produce a false alarm and small enough that an actual regression would be caught before reaching production users.

The throughput numbers translate to roughly 30 frames per second on `input_1` for the algorithm itself, falling to roughly 10 frames per second once full export is enabled. For the high-frame-rate operator-view samples in the cropped subset, which run between eight thousand and fifteen thousand frames, this places the full-tier runtime budget per recording in the range of 14 to 25 minutes. Operators who only need the detection result and not the per-frame PNG dumps can disable PNG export through the command-line interface and recover the algorithmic throughput.

IX. Detector Bootstrap Demonstration

VRIFA is positioned in part as a label-generation engine for downstream learned detectors. To exhibit that pathway concretely, this section reports a single qualitative demonstration in which a YOLO segmentation model [11] is trained from the COCO export produced by VRIFA's annotation pipeline and then applied to a held-out frame from one of the eleven samples. The training set consisted of the COCO export from the canonical reference videos `input_1` and `input_2`, comprising 4, !157 and 205 wet-region annotations respectively. Training was a single short run with default Ultralytics hyperparameters, no manual tuning, and no separate validation set tied to the human-labeled subset described in Section 5. The result is shown in Figure Fig. 8.

YOLO bootstrap placeholder. A held-out frame with the YOLO-predicted wet-region overlay. Replaced once the bootstrap artifact is regenerated against the current Rust default configuration.

Fig. 8 Qualitative result of bootstrapping a YOLO segmentation model from VRIFA's COCO export. The demonstration is intentionally qualitative; quantitative downstream-detector evaluation is deferred to follow-on work with a separately-collected detector-validation set.

The result should be read as a proof-of-concept for the bootstrap pathway, not as a benchmark of the trained detector. There is no quantitative claim about the YOLO model in this paper; in particular, no IoU or precision-recall metric is reported for the detector itself. Two reasons. First, the annotation export from VRIFA is uncalibrated supervision and the trained detector inherits whatever systematic error the upstream classical pipeline produces; reporting the detector's accuracy against the same fifty-five labels would conflate the labeling-pipeline error with the detector-architecture error. Second, a quantitative claim about the trained detector requires a held-out detector-validation set that does not overlap with the training videos, and that set has not been collected. Both gaps are followed up in dedicated work and are out of scope for this submission.

X. Discussion

VRIFA is a working baseline, not a universal segmenter. The failure modes described below are concrete, mechanism-level, and visible from the operator console, which is the same place the central processing unit (CPU) detection pipeline runs in production.

A. Failure Modes

The temporal-locking parameter is the primary diagnostic when results appear noisy on long pauses. The pipeline exposes a temporal-locking window, defaulting to three frames. Locking holds a positive detection in the mask for that many subsequent frames so transient single-frame dropouts do not flicker the boundary. When the operator sets the temporal-locking window higher than the expected pause length in the run, transient detections from earlier in the infusion persist past the moment the front recedes, and the mask grows phantom regions that look like artifacts. The mechanism is not a defect. It is the locking semantics doing exactly what they were configured to do, applied to a sequence whose pauses violate the implicit assumption. The operational consequence is that the temporal-locking window is a per-mold parameter, not a global default, and the failure is visible only on long-pause sequences.

The dynamic-reference calibration window is the second sensitive surface. The pipeline exposes a dynamic-calibration-frames parameter, defaulting to ten, and the dynamic reference mode uses those first ten processed frames to fit a square-root-of-area growth model that subsequently lags the reference frame behind the live frame. The assumption baked into that fit is that early-fill is representative of the growth law. If the first ten processed frames contain race-tracking, an air pocket, or some other anomaly, the fit picks up a poor reference factor and every downstream frame inherits that error. The fix is operator-side, namely choose a calibration window that excludes the anomaly or fall back to one of the static reference modes. The failure is again a misuse of a parameter rather than a numerical defect.

The third failure mode is the only one that is purely a consequence of the artifact pipeline rather than the algorithm. Two encodes of identical raw overlay and heatmap arrays produce MPEG-4 (MP4) streams whose decoded frames differ from the source arrays by a mean per-pixel L2 norm near 5.0 for the overlay video and near 3.5 for the heatmap video. The Portable Network Graphics (PNG) outputs, written through a non-codec path, are byte-exact across runs. The mechanism is libavcodec slice-thread nondeterminism in the encode step, not anything in the core algorithm crate. Because the divergence is real and well-characterized, the parity gate runs on PNGs and the structured run summary, and the MP4s are treated as human-review artifacts that are visually identical but not byte-identical.

The fourth failure mode is the inherent ceiling of any tuned classical-CV pipeline. VRIFA does not generalize across substantially different fabric types, lighting setups, or vacuum-bag textures without re-tuning. The default parameters assume a transparent vacuum bag, top-down LED lighting, and a dark mold background, because those are the conditions of the sample videos in the bundled dataset directory. Off-distribution conditions, such as a heavily textured bag, side lighting, or a bright mold surface, can break the default Otsu threshold or push the response distribution outside the range the threshold offset was tuned for. The pipeline provides a percentile-threshold mode, a manual contrast threshold, and an RGB colorspace switch precisely because the default cannot cover every regime, and switching them is the documented response.

B. Tradeoffs

VRIFA is built on a deliberate set of tradeoffs that a reader should weigh alongside the benchmark numbers.

We trade an end-to-end accuracy ceiling for explainability and per-mold tunability. A learned segmenter trained on a sufficiently large in-domain corpus would likely exceed the boundary F1 reported in this paper. We do not have a sufficiently large in-domain corpus, and a research group that wants to run a vacuum-assisted resin transfer molding (VARTM) experiment next week does not have the time to label one. The Application contribution is precisely the claim that this tradeoff is favorable in the VARTM regime, where samples are small, ground truth is sparse, and the operator already understands the physics. The pipeline runs on a CPU with no graphics processing unit (GPU), which keeps the deployment surface small and matches how VARTM rigs are actually instrumented in academic and industrial labs.

We trade automated hyperparameter selection for human inspection. There is no Optuna-driven default and no opaque end-to-end model. The operator sees what each parameter does. The roughly fifty configuration parameters are a feature for research, where the practitioner needs to know exactly why a particular result looks the way it does, and a hazard in production, where an unfamiliar operator can choose a temporal-locking window that ruins a run. We are explicit about that tension rather than pretending the parameters are zero-cost.

We trade end-to-end byte-exactness across runs for a working video output. The MP4 streams are not bit-reproducible, and we report rather than hide the encode-tier noise. Bit-reproducibility is preserved where it matters for downstream supervision, namely on the PNG outputs and on the COCO annotations, and abandoned where it does not, namely on the human-review video.

We trade closed-source convenience for an auditable open reference implementation. Anyone can recompile the workspace, rerun the benchmark gate, and check their own outputs against the committed reference outputs directory. The Rust source replaces an opaque baseline with one whose every stage is named, modular, and addressable.

C. Scope and Limits

Several setups fall outside the regime VRIFA targets. Opaque vacuum bags break the entire pipeline, because the front is no longer visually observable from above and there is nothing for any image-domain method to detect. Multi-resin systems where dye contrast is intentionally absent fall in the same category, since the response image VRIFA computes is, ultimately, a contrast measurement. Very fast infusions, where the front passes through a region of interest in well under a second, push the temporal-locking and morphology kernels past the regime they were designed for, and the resulting masks lag the true front. Camera motion during infusion is handled by the optional registration stage, which warps the live frame back into the reference coordinate system whenever per-frame or cumulative drift exceeds an operator-set threshold. The registration absorbs the bumped-tripod and thermal-creep regimes that previously broke the comparison between live and reference frame; setups where the camera moves continuously and at high amplitude, such as a handheld phone walked around the rig, exceed the small-shift assumption the affine warp is fit under and remain out of scope.

These are not weaknesses to be patched. They are the boundary of the application domain. A practitioner facing any of them requires a different tool, and stating that boundary explicitly is more useful than overstating the method's operating envelope.

D. Implications for the Field

VRIFA is intended to be useful in three concrete ways. Practitioners running a VARTM rig obtain a working flow-front baseline that runs on the laptop next to the rig and produces overlays, masks, heatmaps, and structured annotations without a labeling campaign. Researchers comparing new methods obtain a public reference implementation pinned to a specific commit, a specific configuration, and a specific set of reference outputs, which means a comparison can be replicated rather than approximated. Downstream-detector trainers obtain a labeled bootstrap set produced from the same binary, in COCO, YOLOv5, and Darknet formats, with deterministic PNG masks and deterministic annotations. VRIFA is sufficient to bootstrap detector training and to serve as a reproducible reference implementation for permeability-inversion and process-monitoring studies, and the artifact in the repository is what makes that claim falsifiable. The tradeoffs above are the price of that falsifiability, and they are appropriate for the regime VRIFA targets.

XI. Conclusion

The integrated pipeline reaches mask IoU $X.XXX$ and boundary F_1 $Y.YYY$ across a 55-frame hand-labeled subset spanning eleven distinct VARTM infusion runs. Removing any single component, namely the peak-brightness reference, the darken-only delta, the region-of-interest restriction, the dynamic-lag reference selection, the morphological cleanup, or the temporal lock, drops mean IoU by at least ΔIoU absolute on the same subset. The pipeline runs at 30 frames per second on a single CPU. The empirical contribution is therefore not any one of the primitives in isolation, all of which already appear in prior VARTM and Liquid Composite Molding work, but the joint configuration and the per-component evidence that the joint matters.

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